Machine Learning

DSA-447/DGE-301

1. Introduction

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Topics

- Course Description
- What is Machine learning
  - Core types of Machine learning
- Data in Machine learning
- Machine learning Lifecycle

The Power of Learning Machines

- Machine Learning is not just about algorithms, it's about teaching computers to learn from experience, just like humans.

💡 **Why it matters:**

- ML powers everything from **Google Search** to **Netflix recommendations**, **medical diagnosis**, and **self-driving cars**.
- In the 21st century, data is the new oil, and ML is the engine that drives innovation.
- What used to require human intuition can now be done by machines — faster, smarter, and at scale.

The best way to predict the future is to create it, with Machine Learning

How Machine Learning Shapes Our World

- Your smartphone unlocks with face recognition → ML
- You watch your favorite movie suggestion → ML
- Your email filters spam automatically → ML
- Doctors predict disease early → ML
- Cars drive themselves → ML

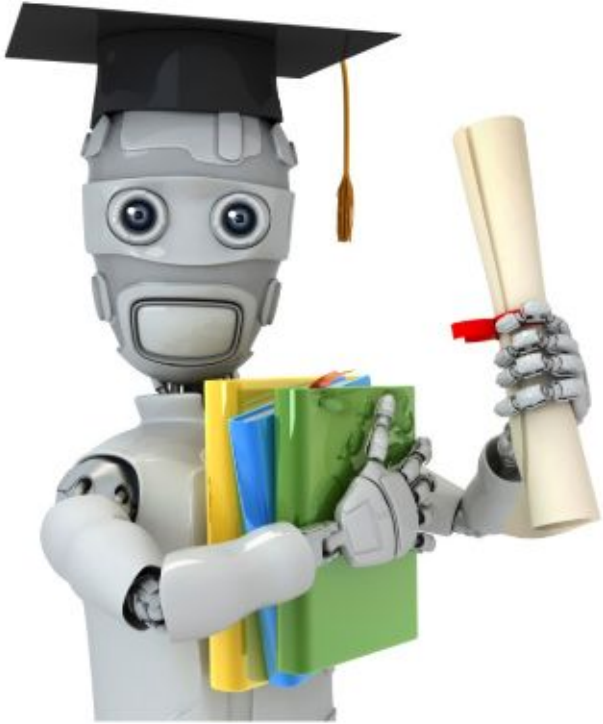
What Will We Learn in This Course?

- In this course you will learn to:
  - Understand what Machine Learning really means.
  - Explore the different types of learning, supervised, unsupervised, and reinforcement.
  - Discover the ML workflow, from data collection to model evaluation.
  - Get hands-on with Python and Scikit-Learn.
  - Build your first ML models step by step.

The full course content will be shared with student in separate file.

Course Policies

- **Grade**
  - 30 % for homeworks
  - 25 % for midterm
  - 25 % for final
  - 20 % for projects
- **Communication and Collaboration**
  - We will use Discord or Line group for instance communication and collaboration.
  - (Please join) https://discord.gg/4DWUQkgt



Machine Learning

Introduction

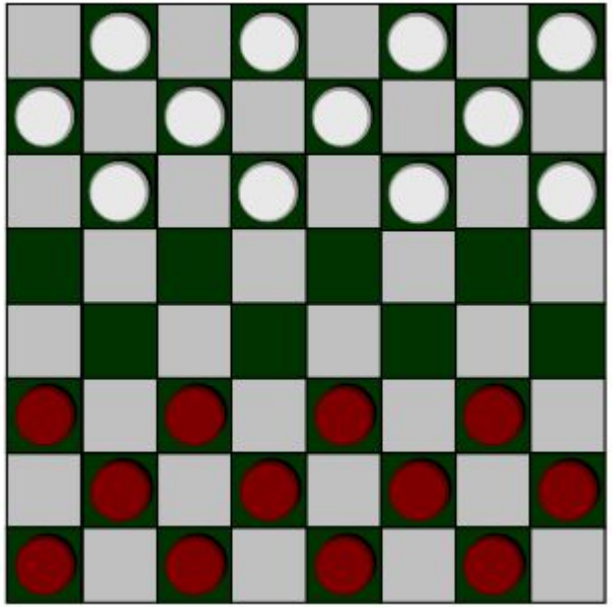
What is machine learning

Definition

- **Arthur Samuel (1959)-** Machine learning is a field of study that gives computers the ability to learn without being explicitly programmed.

Imagine you build a computer program to play **checkers**

1. At first, the computer knows nothing.
2. After each game, it receives feedback (reward or penalty).
3. Over time, it improves.



Definition

- **Tom Mitchell (1998)-** A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. **What is the task T in this setting?**

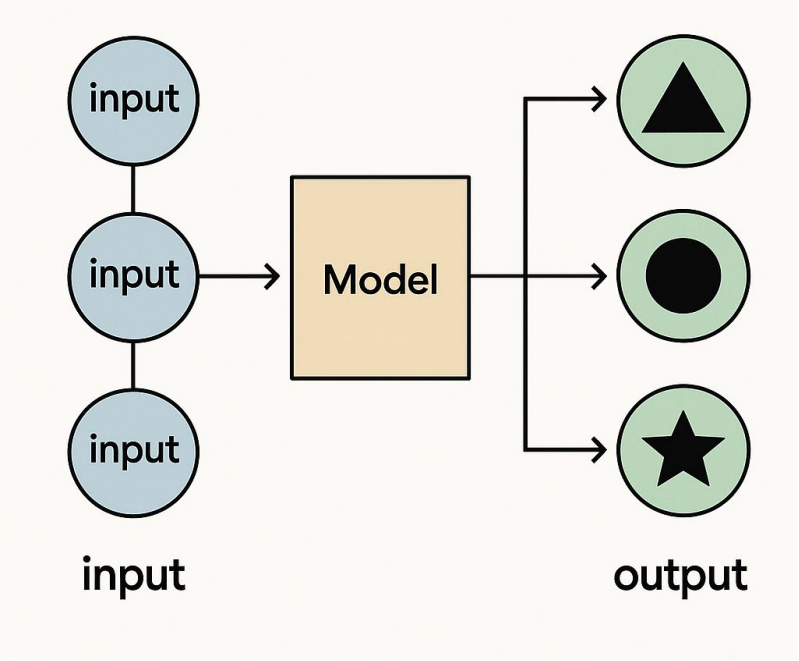
- Classifying emails as spam or not spam.
- Watching you label emails as spam or not spam.
- The number (or fraction) of emails correctly classified as spam/not spam.

Types of Machine Learning

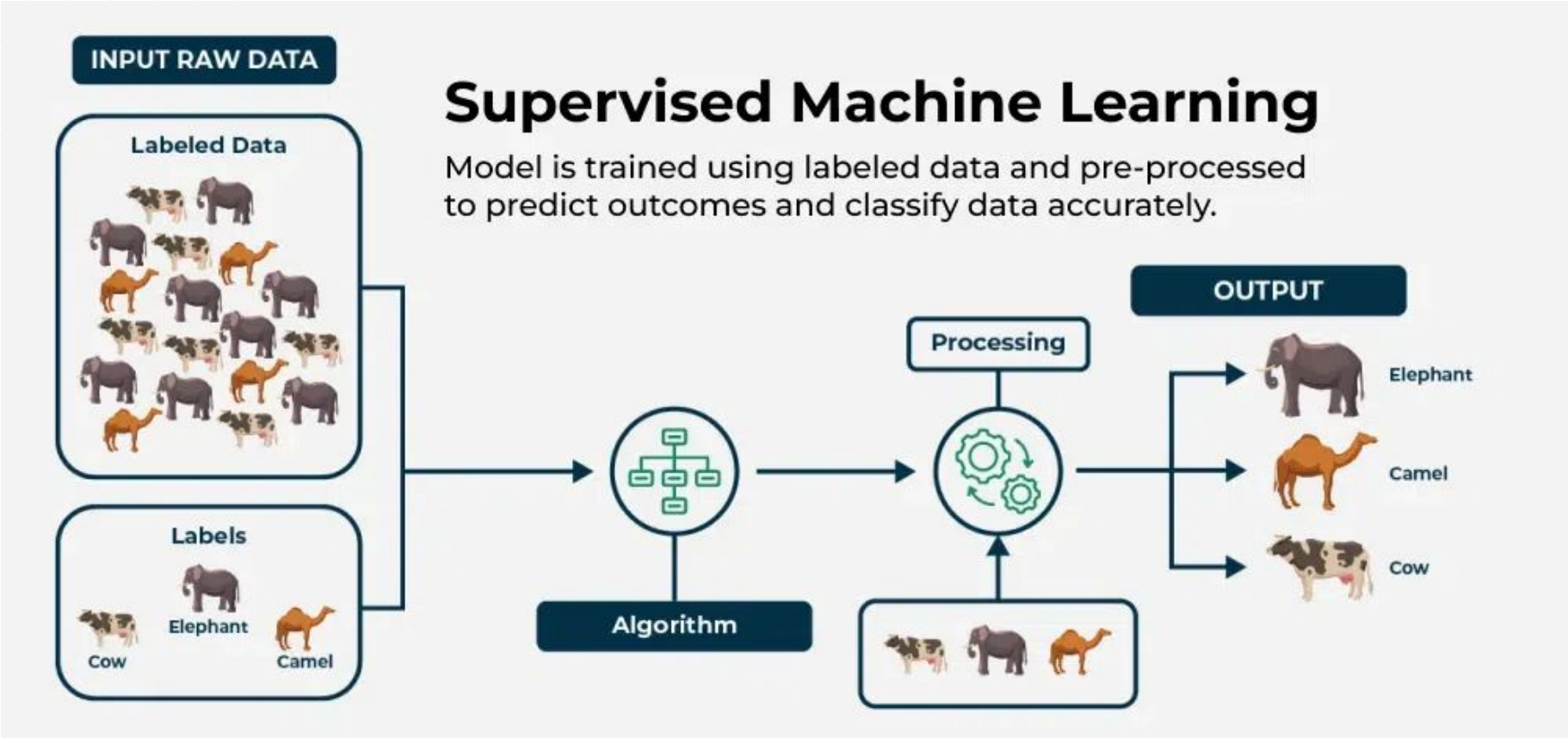
- **Three core types:**
  - Supervised Learning
    - Trains models on labeled data to predict or classify new, unseen data.
  - Unsupervised Learning
    - Finds patterns or groups in unlabeled data, like clustering or dimensionality reduction.
  - Reinforcement Learning
    - Learns through trial and error to maximize rewards, ideal for decision-making tasks.

Types of Machine Learning| Supervised Learning

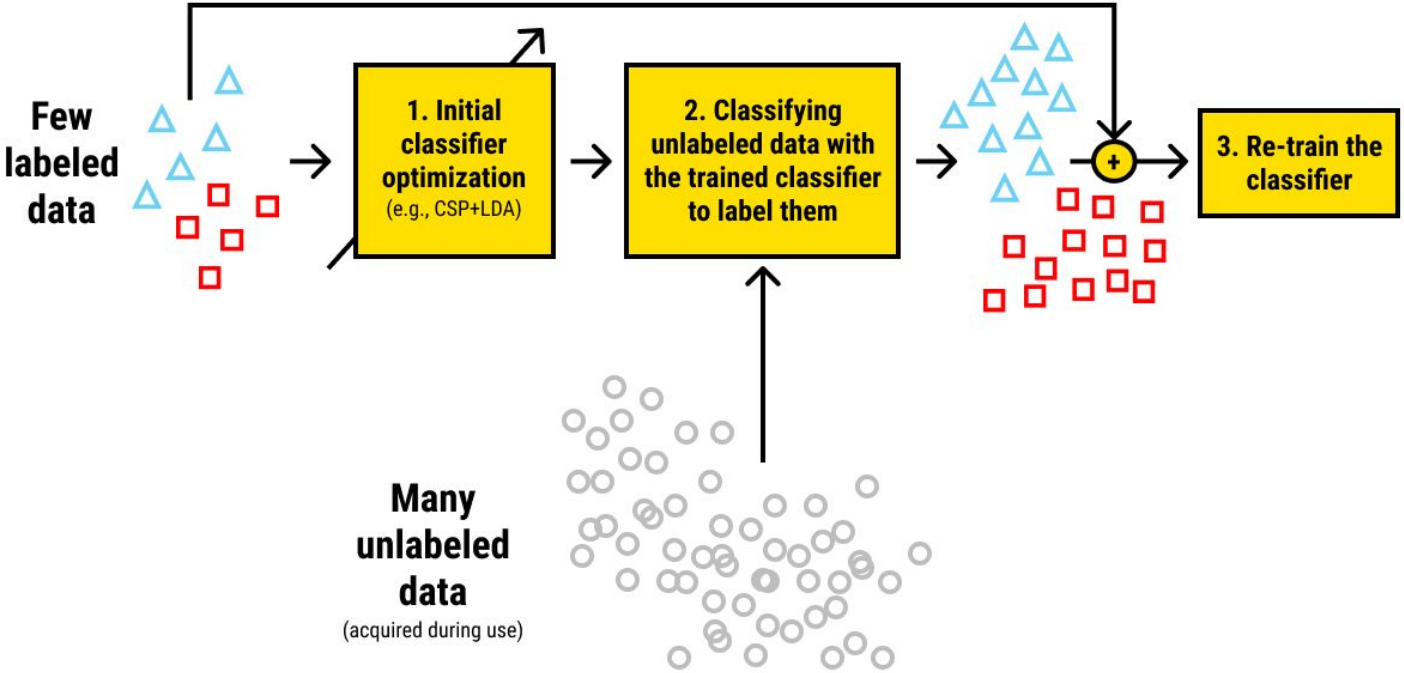
- A model learns from a labelled data (every input has a corresponding correct output).
- The model makes prediction and compares them with the true outputs.
- The goal is to make accurate predictions on new, unseen data.
- for example,
  - A model trained on images of handwritten digits can recognise new digits it has never seen before.



Types of Machine Learning| Supervised Learning



Types of Machine Learning| Supervised Learning



Types of Machine Learning| Supervised Learning

Supervised Machine Learning

Regression

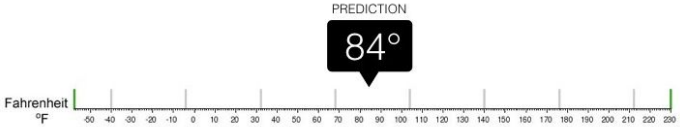
- Predicts continuous numeric value
- Example: Stock price, temperature, etc.
- Common used regressors: Linear regression, random forest regressor, ridge regression, etc.
- Evaluation metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), etc.

Classification

- Predicts discrete categorical value
- Example: Customer chunk, email spam or ham
- Commonly used classifiers: Decision tree classifier, random forest classifier, SVM, etc.
- Evaluation metrics: Accuracy, Precision, Recall, F1 Score, etc.

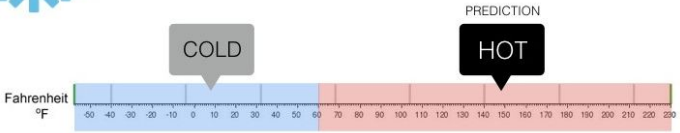
Regression

What is the temperature going to be tomorrow?



Classification

Will it be Cold or Hot tomorrow?



Types of Machine Learning| Supervised Learning

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

Types of Machine Learning| Supervised Learning

Working of Supervised Machine Learning

1. Collect Labeled Data

Gather a dataset where each input has a known correct output (label).

Example: Images of handwritten digits with their actual numbers as labels.

2, Split the Dataset

Divide the data into training data (about 80%) and testing data (about 20%).

The model w°lll learn from the training data and be evalua ted on the testing data

3. Train the Model

Feed the training data (inputs and their labels) to a suitable superv-vised learning algorithm (like Decision Trees, SVM or Linear Regression).

The model tries to find patterns that map inputs to correct outputs

4. Validate and Test the Model

Evaluate the model using testing data it has never seen before.

The model predicts outputs and these predictions are compared with the actual labels to calculate accuracy or error

#	Scenario	Prediction Task	Type of Problem (Classification / Regression)
1	Student Exam Results	Predict whether a student will Pass or Fail	☐
2	Student Exam Results	Predict the Exact Exam Score (out of 100)	☐
3	Medical Health Prediction	Predict whether a patient has diabetes (Yes/No)	☐
4	Medical Health Prediction	Predict the blood sugar level of a patient	☐
5	Housing Market Analysis	Predict whether a house is Affordable, Average, or Luxury	☐
6	Housing Market Analysis	Predict the actual market price of the house	☐
7	Customer Behavior Prediction	Predict whether a customer will buy a product (Yes/No)	☐
8	Customer Behavior Prediction	Predict how much money the customer will spend next month	☐
9	Weather Forecasting	Predict whether it will Rain Tomorrow (Yes/No)/ Predict tomorrow's temperature (°C)	☐/☐

Types of Machine Learning| Supervised Learning

- Supervised Machine Learning Algorithms
  - Linear Regression
  - Logistic Regression
  - Decision Tree
  - Random Forest
  - Support Vector Machine
  - Gradient Boosting
  - Naive Bayes Algorithms.

We will cover all these algorithms during this course.

Types of Machine Learning| Supervised Learning

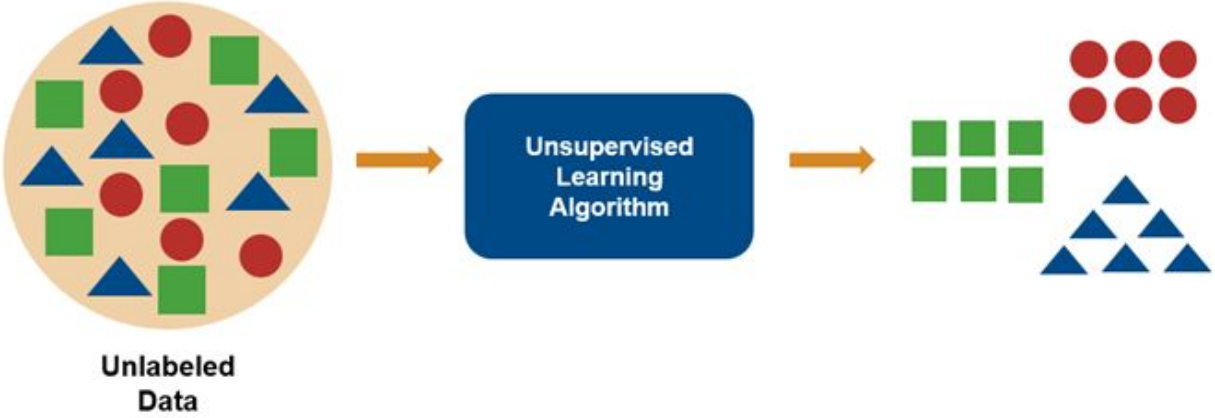
- **Advantages:**
  - Simplicity & clarity: Easy to understand and implement since it learns from labeled examples.
  - High accuracy: When sufficient labeled data is available, models achieve strong predictive performance.
  - Versatility: Works for both classification like spam detection, disease prediction and regression like price forecasting.
  - Generalization: With enough diverse data and proper training, models can generalize well to unseen inputs.
  - Wide application: Used in speech recognition, medical diagnosis, sentiment analysis, fraud detection and more.

Types of Machine Learning| Supervised Learning

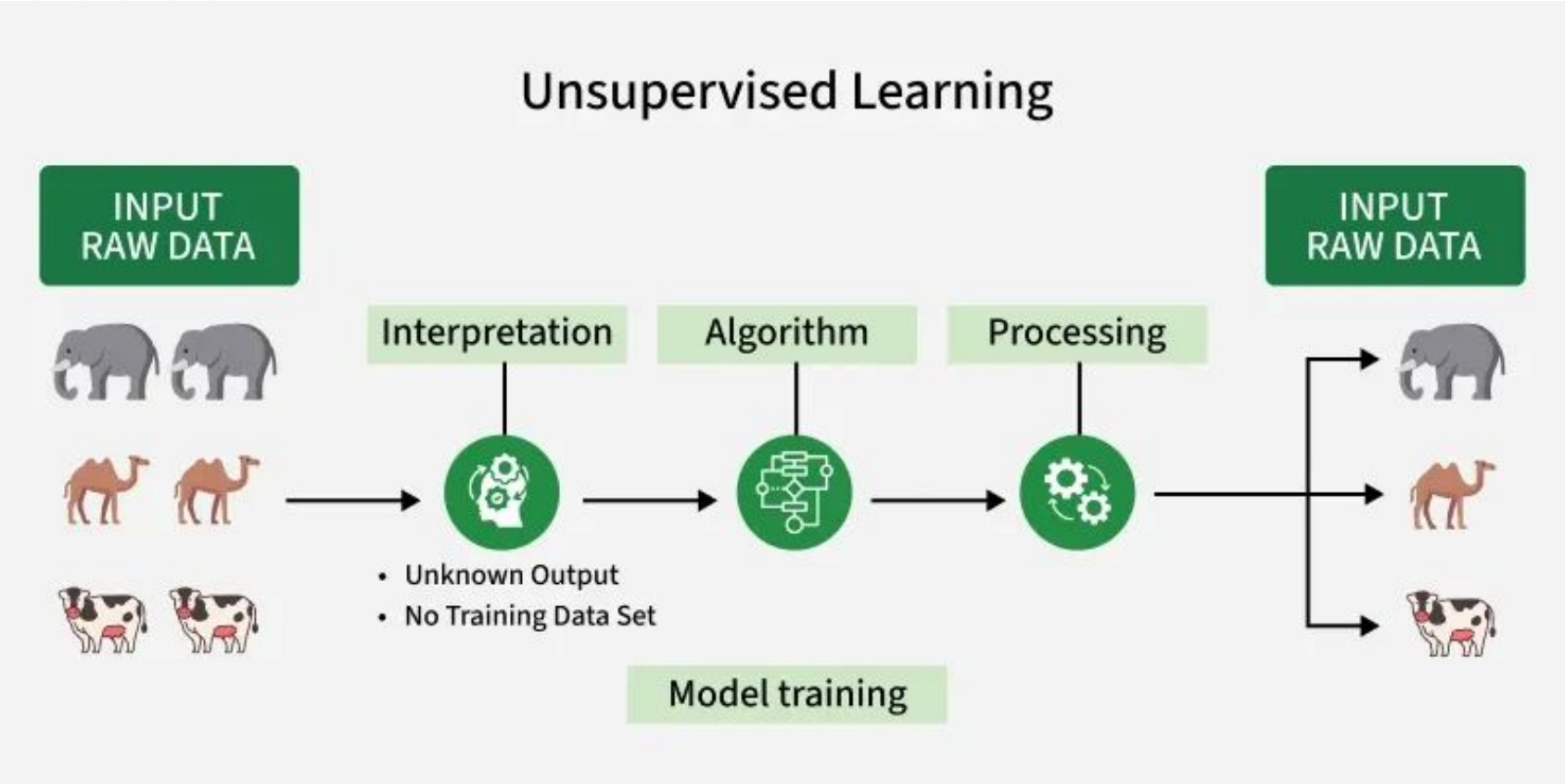
- **Disadvantages:**
  - Requires labeled data: Large amounts of labeled datasets are expensive and time-consuming to prepare.
  - Bias from data: If training data is biased or unbalanced, the model may learn and amplify those biases.
  - Limited adaptability: Performance drops significantly when applied to data distributions very different from training data.
  - Not scalable for some problems: In tasks with millions of possible labels like natural language, supervised labeling becomes impractical.

Types of Machine Learning| Unsupervised Learning

- Models data without labelled responses or predefined categories.
- Unsupervised learning algorithms work solely with input data and aim to discover hidden **patterns**, **structures** or **relationships** within the dataset independently, without any human intervention or prior knowledge of the data's meaning.

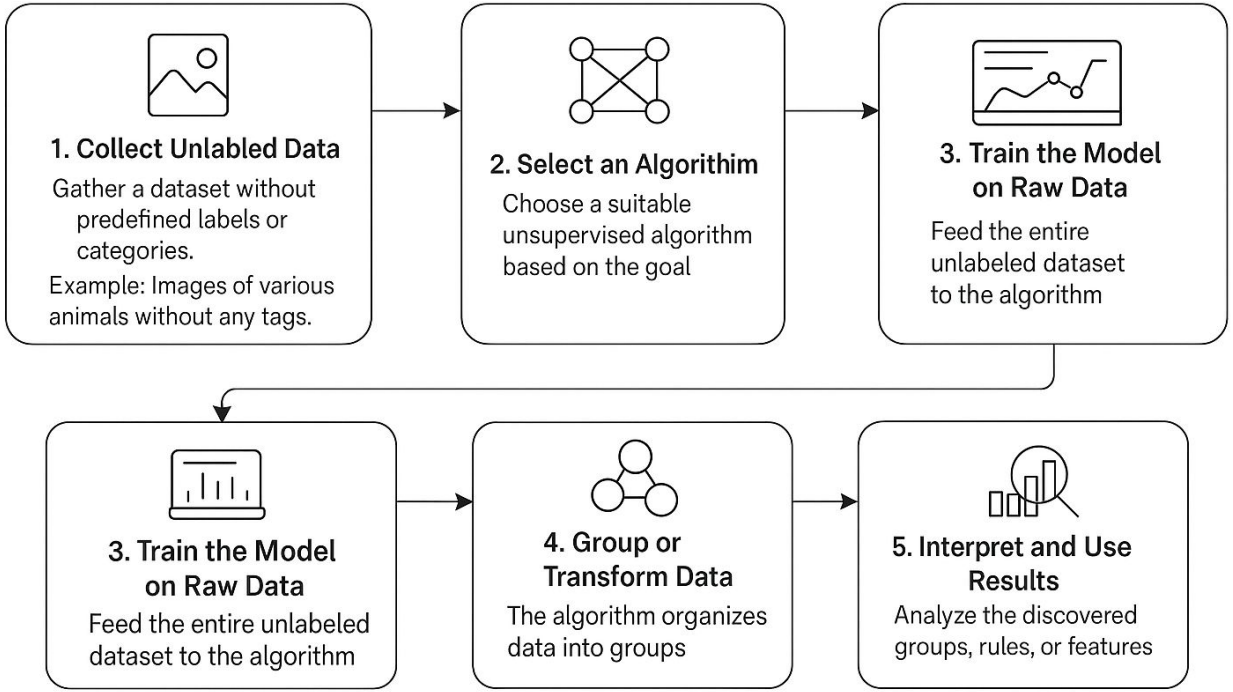


Types of Machine Learning| Unsupervised Learning



Types of Machine Learning| Unsupervised Learning

Working of Unsupervised Learning



Types of Machine Learning| Unsupervised Learning

Types of Unsupervised Learning

1. Clustering Algorithms
   - Groups unlabeled data into clusters based on similarity. Its goal is to discover ***patterns or relationships*** within the data without any prior knowledge of categories or labels.
2. Association Rule Learning
   - a rule-based technique used to discover interesting relationships between variables in large datasets. It identifies patterns in the form of “if-then” rules, showing how the presence of some items in the data implies the presence of others.
3. Dimensionality Reduction
   - The process of decreasing the number of features or variables in a dataset while retaining as much of the original information as possible.

Types of Machine Learning| Unsupervised Learning

Applications of Unsupervised Learning

- Customer Segmentation: Algorithms cluster customers based on purchasing ***behavior*** or ***demographics***, enabling targeted marketing strategies.
- Anomaly Detection: Identifies unusual patterns in data, aiding fraud detection, cybersecurity and equipment failure prevention.
- Recommendation Systems: Suggests products, movies or music by analyzing user behavior and preferences.
- Image and Text Clustering: Groups similar images or documents for tasks like organization, classification or content recommendation.
- Social Network Analysis: Detects communities or trends in user interactions on social media platforms.

Types of Machine Learning| Unsupervised Learning

Advantages

- No need for labeled data: Works with raw, unlabeled data hence saving time and effort on data annotation.
- Discovers hidden patterns: Finds natural groupings and structures that might be missed by humans.
- Handles complex and large datasets: Effective for high-dimensional or vast amounts of data.
- Useful for anomaly detection: Can identify outliers and unusual data points without prior examples.

Types of Machine Learning| Unsupervised Learning

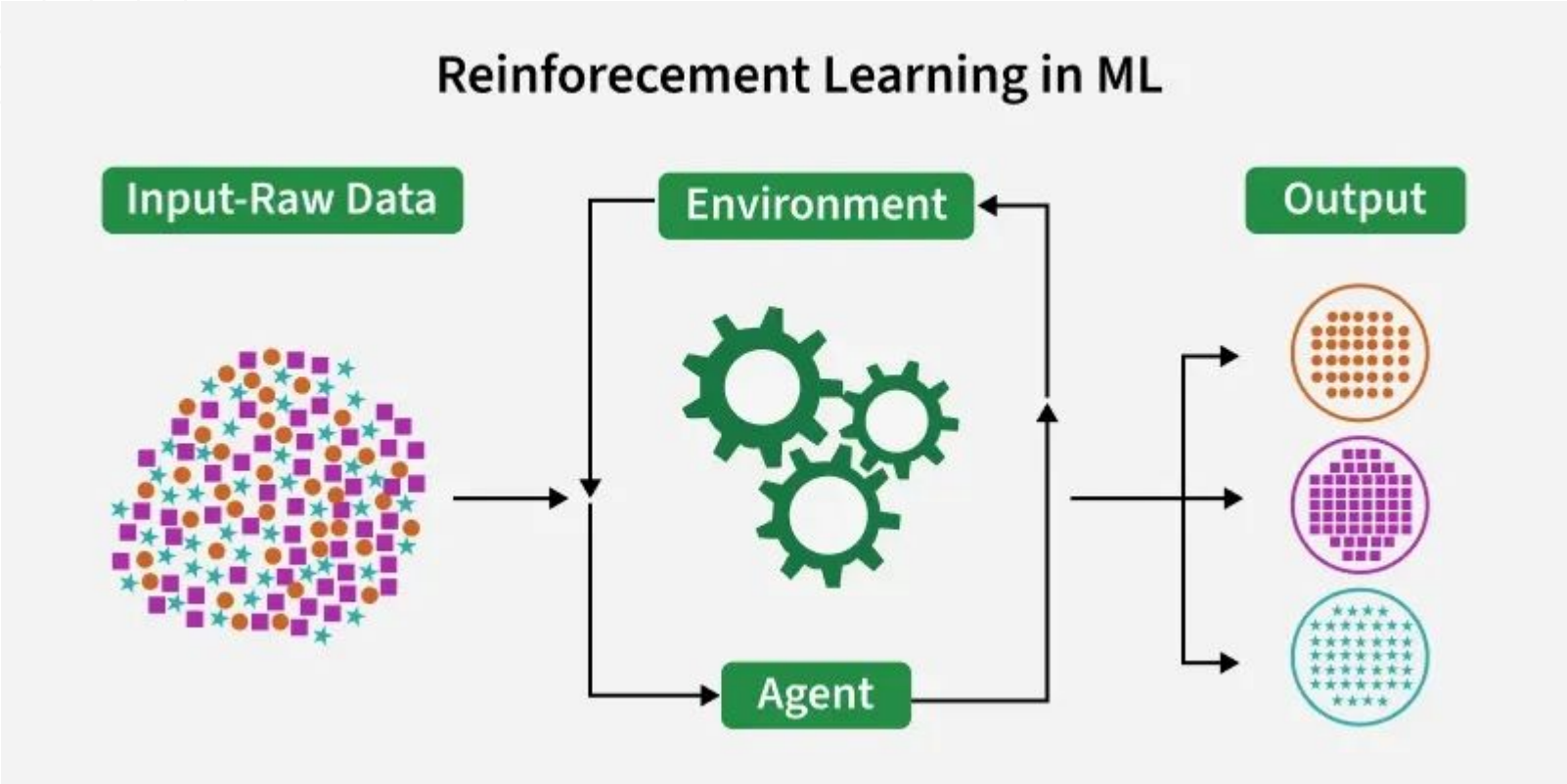
Disadvantages

- Noisy Data: Outliers and noise can distort patterns and reduce the effectiveness of algorithms.
- Limited Guidance: The absence of labels restricts the ability to guide the algorithm toward specific outcomes.
- Cluster Interpretability: Results such as clusters may lack clear meaning or alignment with real-world categories.
- Lack of Ground Truth: Unsupervised learning lacks labeled data making it difficult to evaluate the accuracy of results.

Types of Machine Learning| Reinforcement Learning

- Focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards.
- Allows machines to learn by interacting with an environment and receiving feedback based on their actions.
- This feedback comes in the form of rewards or penalties.
- Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal.
- The agent performs actions and receives feedback to optimize its decision-making over time.

Types of Machine Learning| Reinforcement Learning



Types of Machine Learning| Reinforcement Learning

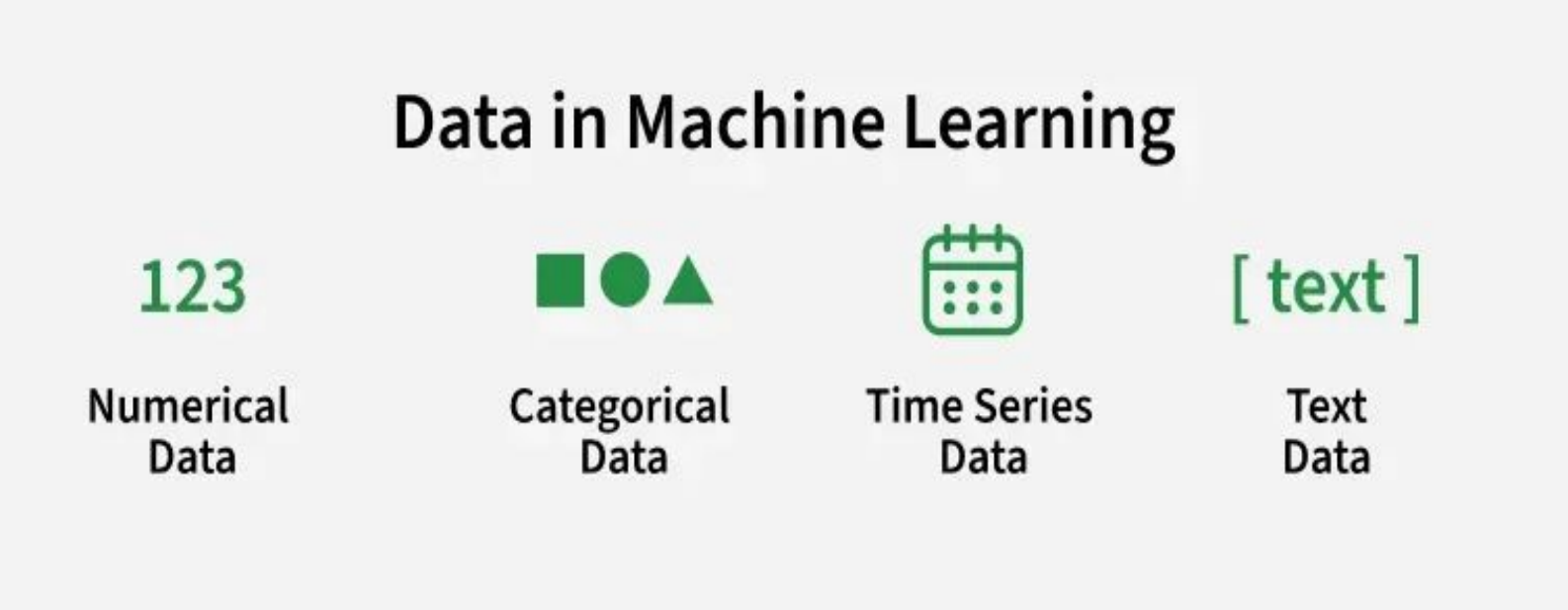
- **Agent**: The decision-maker that performs actions.
- **Environment**: The world or system in which the agent operates.
- **State**: The situation or condition the agent is currently in.
- **Action**: The possible moves or decisions the agent can make.
- **Reward**: The feedback or result from the environment based on the agent's action.

Data in Machine Learning

Data

- Data refers to the set of observations or measurements to train a machine learning models.
- The performance of such models is heavily influenced by both the quality and quantity of data available for training and testing.
- Machine learning algorithms cannot be trained without data. Cutting-edge development in Artificial Intelligence, automation, and data analysis is powered mostly by vast sets of data.
- For example, Facebook acquired WhatsApp for $19 billion primarily to access user data, which is critical for enhancing services.

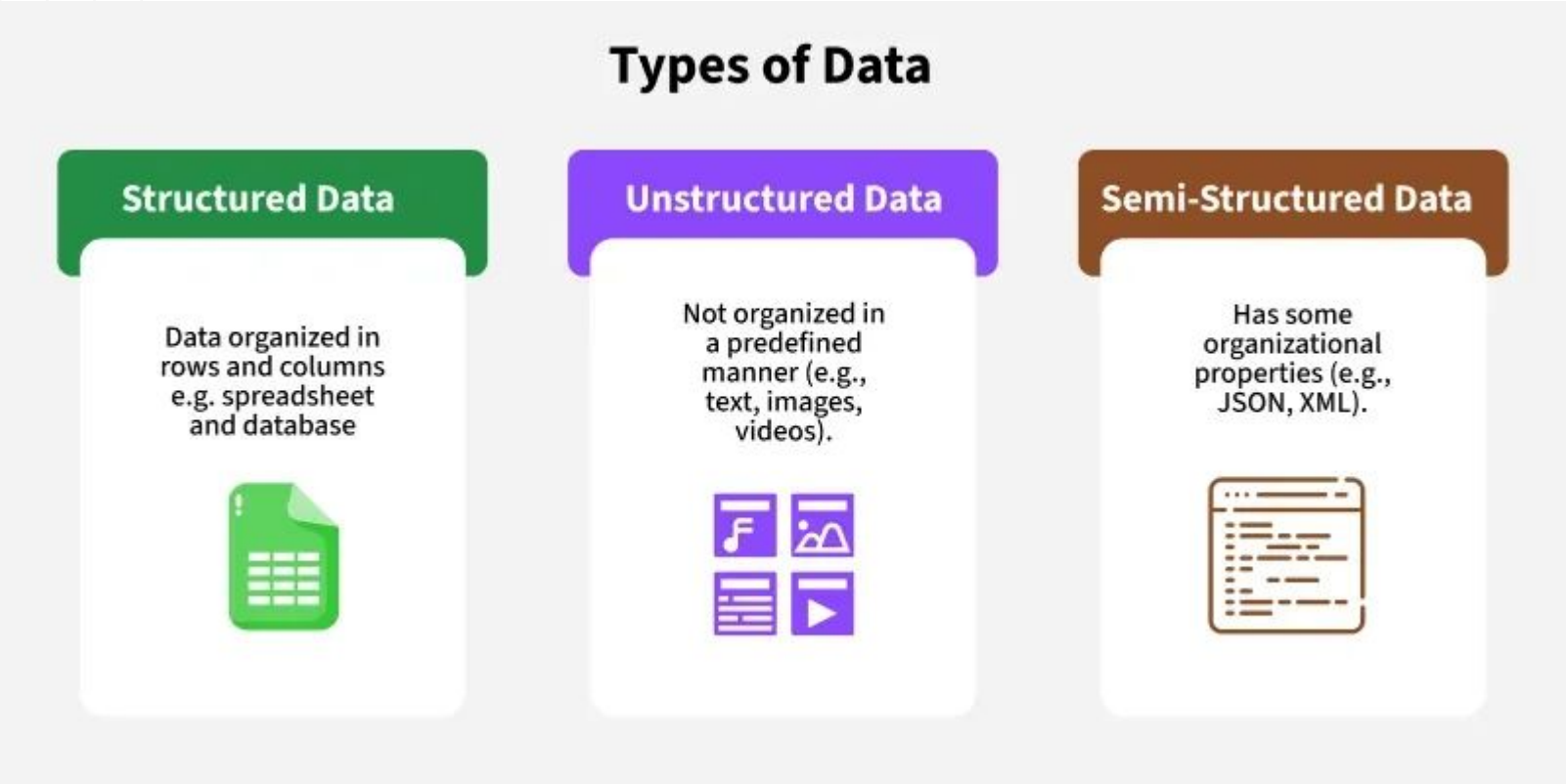
Data


Data in Machine Learning
123
Numerical Data
Categorical Data
Time Series Data
[text]
Text Data

Data

- **Characteristics:**
  - Volume: The scale of data generated every millisecond.
  - Variety: Different data types like healthcare, images, videos, and audio.
  - Velocity: The speed of data generation and streaming.
  - Value: The meaningful insights data provides.
  - Veracity: The accuracy and reliability of data.
  - Viability: Data's adaptability for integration into systems.
  - Security: Preventing tampering and unwanted access.
  - Accessibility: Simple access for decision-making.
  - Integrity: Accuracy and consistency throughout its lifecycle.
  - Usability: Simplicity and interpretability for end-users.

Types of Data in Machine Learning



Types of Data in Machine Learning

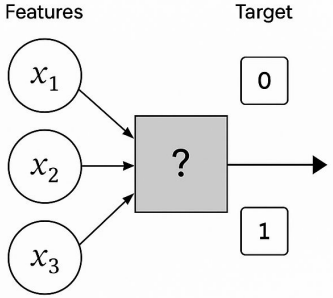
- **Based on Representation**
  1. Numerical Data: Features measured in numbers (e.g., age, income).
  2. Categorical Data: Represents Categories or labels (e.g., gender, fruit type).
  3. Ordinal Data: Categorical data with an essential order (e.g., clothing sizes: Small, Medium, Large).
- **Based on Labeling**
  1. Labeled Data: Includes input variables and corresponding target outputs. Example: Features like "age" and "income" with a label like "loan approval status."
  2. Unlabeled Data: Contains only input variables without any target labels. Example: Images without annotations.

Types of Data in Machine Learning

- **Measurements on different aspects of n objects:**
  1. **Target**: output variable / goal of prediction (what we want the model to predict or learn from those inputs)
  2. **Features**: properties that describe the object (input variables, the information we give to the model)

We assume some relationship between features and target, and want to discover that predictive rule.

Features and Target



Predicting House Prices

Size (sq.ft)	Bedrooms	Location Score	Age (years)	Price (USD)
1200	3	8,2	10	220.000
850	2	6,5	15	150.000
1900	4	9,1	5	310.000
1500	3	7,8	8	250.000

From Data to Knowledge

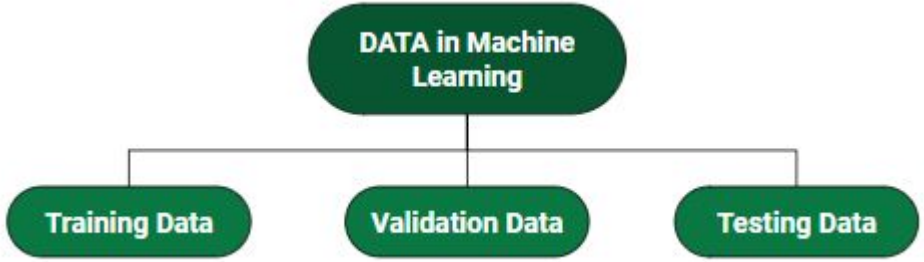
- **Data:** Data is raw, unprocessed facts, values, text, sounds or images that have not been interpreted or analyzed. Without data, training models and driving modern research or automation would be impossible.
- **Information:** As data gets processed, interpreted and organized, it turn into information. It gives users meaningful insights which can be understood easily and utilized.
- **Knowledge:** Knowledge is the product of combining experience, learning, Information and insights. It allows individuals or businesses to construct awareness, create ideas and make well-informed decisions.

From Data to Knowledge

Example:

A store collects customer feedback (raw data). Analyzing this data for common themes (e.g., product quality, pricing) creates information. Applying these insights to improve product offerings results in knowledge.

How do we split data in Machine Learning?



1. Training Data: Used to train the model.

2. Validation Data: Helps fine-tune the model by evaluating it during training.

3. Testing Data: Used after training is complete. Evaluates how well the model generalizes to unseen data

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